

Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,242 earthquake events in Myanmar from 1970-07-29 to 2026-04-14. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 0.350 with magnitude of completeness $M_c=2.4$.

Total Events: 9,242

Date Range: 1970-07-29 to 2026-04-14

Max Magnitude: M7.7

Mean Depth: 24.5 km

b-value: 0.350

a-value: 4.685

M_c (completeness): 2.4

2. Earthquake Map

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3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 25 dams

High Risk: 119 dams

Moderate Risk: 100 dams

Low Risk: 10 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist	Fault	Risk	Grade			
Dam near 22.09N 95.86E		0.1658	6.2 km	8.3	Critical				
Dam near 21.77N 95.89E		0.1908	2.9 km	8.3	Critical				
Natkar	0.1508	8.4 km	8.2	Critical					
Meikhtila	0.1483	8.8 km	8.2	Critical					
Ta Nai Hka	Ta Nai Hka	0.1806	4.3 km	8.2	Critical				
Dam near 20.60N 95.90E		0.1540	7.9 km	8.2	Critical				
Dam near 21.74N 95.85E		0.1537	7.9 km	8.2	Critical				
Taungnyo	Taungnyo	0.1999	1.3 km	8.2	Critical				
Dam near 20.85N 96.03E		0.1478	8.9 km	8.2	Critical				
Dam near 21.47N 95.87E		0.1642	6.4 km	8.2	Critical				
Dam near 20.72N 95.90E		0.1674	6.0 km	8.2	Critical				
Dam near 20.47N 96.08E		0.1386	10.6 km	8.1	Critical				
Dam near 17.55N 96.37E		0.1415	10.0 km	8.1	Critical				
Dam near 21.05N 95.83E		0.1327	11.8 km	7.8	Critical				
Dam near 21.04N 95.83E		0.1334	11.6 km	7.8	Critical				
Thaphan Chaung	Thaphan Chaung	0.1305	12.3 km	7.7	Critical				
Kun Chaung	Kun Chaung	0.2009	1.0 km	7.6	Critical				
Dam near 21.34N 95.81E		0.1270	13.1 km	7.5	Critical				
Phyu Chaung	Pyu Chaung	0.1996	1.3 km	7.5	Critical				
Kabaung	Kabaung Chaung	0.1429	9.8 km	7.5	Critical				
Yenwe	Ye Nwe Chaung	0.1821	4.1 km	7.4	Critical				
Nyaunggone		0.1210	14.7 km	7.3	Critical				
Dam near 20.73N 95.83E		0.1217	14.5 km	7.3	Critical				
Dam near 21.37N 95.80E		0.1222	14.3 km	7.3	Critical				
Zeedaw	Nyaung Oat	0.1150	16.4 km	7.1	Critical				
Male-Nattaung	Male-Nattaung	0.1048	20.2 km	6.8	High				
Kyee Ni		0.1036	20.7 km	6.7	High				
Kintat	Mu	0.0990	22.8 km	6.6	High				
Dam near 20.58N 96.17E		0.1005	22.1 km	6.6	High				
Myotha	Ku Htaung	0.1004	22.1 km	6.6	High				

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5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 0.96

c (time offset): 0.119 hours

p (decay exponent): 0.831

Expected M $\geq$ 3 (7d): 0

Expected M $\geq$ 3 (30d): 0

Expected M $\geq$ 3 (90d): 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Boore & Joyner (1997) attenuation model
- Report generated by mmeq-opendata toolkit